

Differential Equations: Formula Sheet & Reference

Class 12 Mathematics – Chapter 9 Complete Reference & Formulas

1 Basic Concepts, Order, Degree & Solutions

1. Differential Equation Definitions:

- **Ordinary Differential Equation (ODE):** An equation involving the independent variable x , dependent variable y , and derivatives of y with respect to x ($y', y'', \dots, y^{(n)}$).
- **Order:** The order of the highest-order derivative appearing in the differential equation.
- **Degree:** The highest power (positive integer) of the highest-order derivative involved, provided the equation is expressible as a polynomial in derivatives ($y', y'', \dots$). If the derivatives are within transcendental functions (e.g., $\sin(y')$, $e^{y'}$), the degree is **not defined**.

2. General vs. Particular Solutions:

- **General Solution (Primitive):** The solution containing as many independent arbitrary constants as the order of the differential equation (n arbitrary constants for order n).
- **Particular Solution:** A solution obtained by assigning specific numerical values to the arbitrary constants (contains 0 arbitrary constants).

2 Method 1: Variable Separable Differential Equations

1. Standard Form & Integration Technique:

$$\frac{dy}{dx} = g(x) \cdot h(y) \implies \frac{1}{h(y)} dy = g(x) dx \quad (h(y) \neq 0)$$

Integrating both sides yields the general solution:

$$\int \frac{1}{h(y)} dy = \int g(x) dx + C$$

2. Exponential Growth & Decay Models:

$$\frac{dP}{dt} = kP \implies \int \frac{dP}{P} = \int k dt \implies P(t) = P_0 e^{kt}$$

3 Method 2: Homogeneous Differential Equations

1. Definition of Homogeneous Functions & Equations:

- A function $F(x, y)$ is homogeneous of degree n if $F(\lambda x, \lambda y) = \lambda^n F(x, y)$ for any $\lambda \neq 0$.
- An equation $\frac{dy}{dx} = F(x, y)$ is homogeneous if $F(x, y)$ is homogeneous of degree zero, i.e., $F(x, y) = g\left(\frac{y}{x}\right)$.

2. Solving $\frac{dy}{dx} = g\left(\frac{y}{x}\right)$:

$$\text{Substitute: } y = v \cdot x \implies \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = g(v) \implies x \frac{dv}{dx} = g(v) - v \implies \frac{dv}{g(v) - v} = \frac{dx}{x}$$

$$\text{Integrate: } \int \frac{dv}{g(v) - v} = \int \frac{dx}{x} + C \quad \left(\text{replace } v \text{ by } \frac{y}{x}\right)$$

3. Solving $\frac{dx}{dy} = h\left(\frac{x}{y}\right)$:

$$\text{Substitute: } x = v \cdot y \implies \frac{dx}{dy} = v + y \frac{dv}{dy}$$

$$\text{Integrate: } \int \frac{dv}{h(v) - v} = \int \frac{dy}{y} + C \quad \left(\text{replace } v \text{ by } \frac{x}{y}\right)$$

4 Method 3: First-Order Linear Differential Equations

1. Type 1: Linear in y ($\frac{dy}{dx} + Py = Q$):

- P and Q are constants or continuous functions of x only.
- **Integrating Factor (I.F.):**

$$\text{I.F.} = e^{\int P dx}$$

- **General Solution Equation:**

$$y \cdot (\text{I.F.}) = \int (Q \cdot \text{I.F.}) dx + C \implies y \cdot e^{\int P dx} = \int (Q \cdot e^{\int P dx}) dx + C$$

2. Type 2: Linear in x ($\frac{dx}{dy} + P_1x = Q_1$):

- P_1 and Q_1 are constants or continuous functions of y only.
- **Integrating Factor (I.F.):**

$$\text{I.F.} = e^{\int P_1 dy}$$

- **General Solution Equation:**

$$x \cdot (\text{I.F.}) = \int (Q_1 \cdot \text{I.F.}) dy + C \implies x \cdot e^{\int P_1 dy} = \int (Q_1 \cdot e^{\int P_1 dy}) dy + C$$